



Application of YOLO deep algorithm in detection of crop diseases and insect pests

Jieming Gao

Tianjin University, Tianjin, 300072, China

Jieminggao975@gmail.com

ABSTRACT

Crop diseases and insect pests are one of the main causes of crop yield and quality decline, causing huge losses to agricultural production. Timely and accurate detection and identification of pests and diseases is critical to taking effective control measures, while traditional manual detection methods are inefficient and error-prone. Using the YOLO algorithm to detect crop diseases and pests has the advantages of high efficiency and accuracy, real-time monitoring, diversified identification, and application automation. Based on the above content, this paper further developed a PestLite-YOLO model for crop pest detectors with various types and intensive detection based on the YOLOv5 model. The above model uses a multi-level pyramid model to extract information in a multi-scale range. At the same time, the mAP50 of the detected sample has been greatly improved, and the final result can be visualized. It is expected to achieve practical application in the detection of crop diseases and insect pests in the future.

CCS CONCEPTS

• **Computing methodologies** → Artificial intelligence; Computer vision.

KEYWORDS

PestLite-YOLO model, image, crop, pest and disease detection

ACM Reference Format:

Jieming Gao. 2024. Application of YOLO deep algorithm in detection of crop diseases and insect pests. In *5th International Conference on Computer Information and Big Data Applications (CIBDA 2024)*, April 26–28, 2024, Wuhan, China. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3671151.3671343>

1 INTRODUCTION

In recent years, efficient and rapid detection of crop diseases and insect pests has become very urgent [1]. Crop diseases and insect pests can cause serious harm to agricultural production and may lead to large-scale crop yield reductions or even harvest failures [2]. Current general detection methods include manual observation, traditional image processing methods and traditional machine learning algorithms [3]. However, these methods have some drawbacks.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

CIBDA 2024, April 26–28, 2024, Wuhan, China

© 2024 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 979-8-4007-1810-6/24/04

<https://doi.org/10.1145/3671151.3671343>

For example, the manual observation method relies on manual visual inspection, which requires a lot of human resources and time [4], and is easily affected by subjective factors and human errors, resulting in low detection efficiency. Traditional image processing methods usually require manual feature extraction and are sensitive to factors such as illumination and angle. They have limited effect on complex scenes and types of pests and diseases. Traditional machine learning algorithms: require a large amount of manual annotation of data and feature engineering, and the processing speed of large-scale data is slow, which is not suitable for real-time detection requirements [5].

YOLO (You Only Look Once) is a popular object detection algorithm that uses deep learning to accurately detect objects in images or videos. In the field of crop disease and pest detection, the application of YOLO is of great significance. Applying the YOLO algorithm to crop disease and insect pest detection has the following advantages and significance [6]. YOLO has the advantages of efficient and rapid detection, automated processing, improved identification accuracy, and promotion of sustainable agricultural development. It can quickly and accurately identify crop diseases and pests in images or videos, achieve real-time detection, detect problems in time and take measures [7]. There is no need to manually extract features, and it has end-to-end detection capabilities, which can realize automated processing, reduce labor costs, and improve detection efficiency. Through deep learning network training and learning features on a large amount of data, different types of pests and diseases can be more accurately identified and the misrecognition rate reduced. The pest and disease data detected based on the YOLO algorithm can be analyzed in combination with other agricultural data to provide farmers with intelligent decision-making support and help them formulate more scientific and effective prevention and control strategies. By improving the efficiency and accuracy of pest and disease detection, the YOLO algorithm helps reduce crop losses and improve agricultural production efficiency, thereby promoting sustainable agricultural development and achieving a more efficient and environmentally friendly agricultural production model.

The PestLite-YOLO model is a special-purpose model based on the YOLO (You Only Look Once) algorithm, specially designed for crop pest and disease detection. The PestLite-YOLO model is based on the YOLO algorithm and uses deep learning technology to achieve rapid and accurate detection of crop diseases and pests [8]. At the same time, the model architecture was optimized for crop pest and disease detection tasks to improve detection efficiency and accuracy. At the same time, the model has been trained on a large dataset of crop pest and disease images, enabling it to identify various types of diseases and pests, including specific problems for different crops. The above model has high real-time performance,

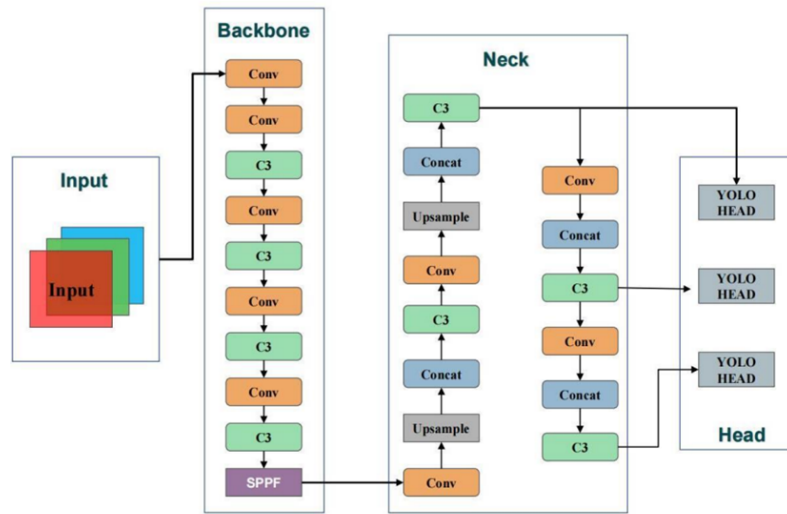


Figure 1: YOLOv5 structural model

can comprehensively detect farmland in a short time, and can detect multiple diseases and insect pests at the same time, improving detection efficiency [9].

Generally speaking, the PestLite-YOLO model can accurately identify various types of crop diseases and pests, including disease spots, insect food, leaf damage, etc., helping farmers to take targeted prevention and control measures in a timely manner; through timely discovery and identification of crop diseases and pests, PestLite- The YOLO model can help farmers take effective control measures to protect crops from pests and diseases, thereby increasing yield and quality; its automated detection function can reduce human resources and time costs, while reducing the use of pesticides and other control materials, reducing production costs cost [10].

In summary, the application of the PestLite-YOLO model in the detection of crop diseases and insect pests is of great significance. It can not only help improve crop yield and quality, but also reduce production costs and promote the sustainable development of agriculture.

Based on the above content, this article proposes a PestLite-YOLO model. The above model uses a multi-level pyramid model to extract information in a multi-scale range. At the same time, the Map50 of the detection sample has been greatly improved, and the final result has been visualized.

1.1 YOLOv5 structural model

YOLOv5 can adapt to image inputs of multiple resolutions and adjust accordingly according to different environments. Its configuration parameters include the number of categories, model depth, and channel multiplier settings for each layer. In addition, in order to adapt to different feature scales, it will also establish multiple anchor point clusters. The backbone of YOLOv5 is built from a set of convolutional layers, mainly used for image feature extraction. These convolutional layers can also capture a variety of features from elementary to complex. First, the dimension of the feature

map is halved by the convolutional layer, then the channel count is gradually increased, and the features are merged through the C3 module (the core unit of YOLOv5). Rewritten results: Finally, the head part uses comprehensive object classification and localization features. First, the features are refined through convolutional layers and then fused with the C3 module. In order to adapt to multi-scale detection, the head also integrates three different output layers, and the structure is shown in Figure 1. Finally, the Detect module represents the converted features as the coordinates, classification and confidence score of the target, thereby ensuring YOLOv5's ability to achieve accurate target detection in various scenarios.

1.2 PestLite-YOLO model structure

In order to meet the detection of huge numbers of pests, we improved the algorithm performance based on the above framework. At the same time, a new model of PestLite-YOLO model structure is proposed, named PestLite-YOLO model. The above model incorporates some relatively novel structural modules and new technologies, and can improve the detection of high-density pests and improve its real-time application. In order to meet the needs of a large number of pest detections, we have improved the performance of the algorithm according to the above framework. At the same time, a new model of PestLite-YOLO model structure was proposed, which was named PestLite-YOLO model. The model is modified on the basis of the YOLOv5 model, retaining the backbone and head of the original model, while adding some innovative technologies and modules. With the introduction of C3ECA, CARAFE, Involution and MTSPPF, the introduction of C3ECA and especially MTSPPF makes the model suitable for the detection of complex multi-scale pests. Involution adaptively adjusts the field of view, which enhances the detection accuracy of microscopic pests. The CARAFE operator helps to retain the feature information during the acquisition process. The introduction of the pyramid model can capture more accurate and detailed information at multiple scales.

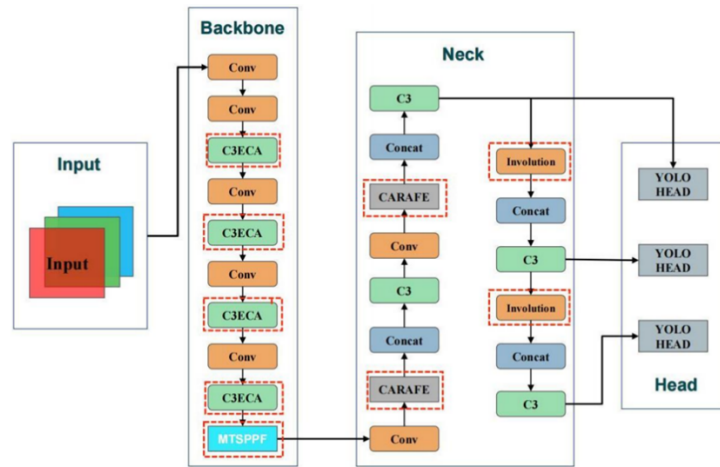


Figure 2: PestLite-YOLO model structure

After improvement, the model effectively solved the problems in high-density pest detection scenarios and significantly improved real-time detection efficiency. By optimizing the algorithm, PestLite can still achieve higher detection accuracy even if the number of parameters is reduced. Therefore, PestLite has become an ideal choice for high-precision real-time pest detection needs in agricultural production, providing an efficient solution for this. The model structure is shown in Figure 2.

1.3 Collection of data sets

Using Mydataset as a data set, it is mainly used to detect pests in crops such as corn and rice. In addition to the "Teddy Cup Pest Dataset", we also captured 600 other images using a Sony Alpha a6400 camera. In order to evaluate and train the learning model of the detector model, maintain the unity and consistency of the data. We selected 1500 images and photos among them for testing.

This dataset mainly contains nine pest species, one of which is rice. Rice plants will gnaw at the rice plants, causing significant damage to the rice plants, which may lead to a significant reduction in rice yield. *Spodoptera litura* is a widespread pest that affects a variety of crops. "Black and white" can eat the leaves of a variety of crops, while "rice blast virus" mainly harms rice, potentially reducing yields. The main insect pest of corn is the "YuMi" creature. The organism's larvae penetrate corn stalks and adversely affect the growing process of the corn. In addition, there is an insect called "Yangxue" that can destroy fruit trees in large numbers. The underground pest 'LouGu' eats away at the roots of plants, causing them to die. At the same time, "Golden Osmanthus" is a multi-functional pest. Its larvae are mainly distributed in the soil, while adults eat leaves and fruits on the ground over a large area. The pest sample image is shown in Figure 3.

2 EXPERIMENTAL CONDITIONS

We mainly ensure the repeatability and reliability of experiments by strictly controlling the following parameters.

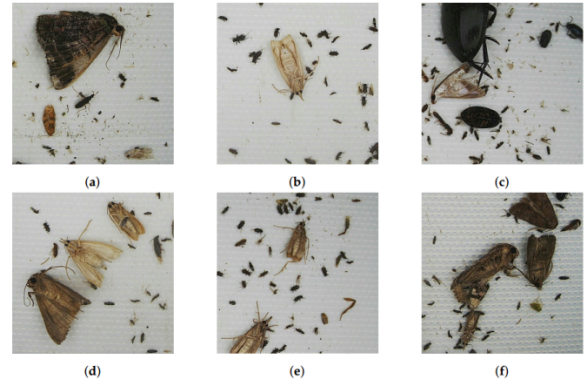


Figure 3: Collect images of pest samples in the library

2.1 Experimental environment

The experiment in this article uses the operating system Ubuntu 20.04. The system's Intel Xeon Gold 6330 processor can accelerate the driver. At the same time, in order to further speed up the calculation, we also installed an NVIDIA RTX 3090 graphics card. The software running is mainly Python 3.9, using PyTorch 1.10.0 version to control and form the entire neural network model. In addition, CUDA 11.4 is used to accelerate the operation of the CPU.

2.2 Input image specifications

Throughout the experiment, the size specification was 640×640 . We used a total of 34 batches and trained the model for a total of 520 rounds. The mosaic method was used for data augmentation to improve model generalization.

2.3 Optimization of learning models

In improving the performance of deep learning models, we use stochastic gradient descent (SGD) as the main optimization method. In order to enhance the efficiency and stability of the parameter update process, we set the momentum parameter to 0.899. The above

method not only effectively accelerates the convergence speed of stochastic gradient descent (SGD), but also reduces oscillation and instability in the parameter space.

In addition, this work also adopts the method of dynamically setting the learning rate to improve the generalization performance of the model for collection of different data sets. Start by setting the learning rate to 1×10^{-2} . Through the continuous decrease of the learning rate during the experiment, it was finally adjusted to 1×10^{-4} to improve the efficiency and stability of the learning model. In addition to the above optimization methods, we also added some data enhancement techniques, such as image flipping, panning and scaling.

The above technologies are used to simulate various changes in the real environment, thereby improving the model's adaptability and robustness to new scenarios.

2.4 Data Accuracy Parametric Model

We use average accuracy (AP), average accuracy (mAP), and number of model parameters (Params). AP is a performance evaluation index of a single category in object detection, and mAP is the average value of multiple categories of aps, which is used to comprehensively evaluate the performance of the overall object detection model. The number of model parameters (params) is used to measure the size of the model. The AP and mAP formulas are as follows:

$$P = \frac{TP}{TP + FP}$$

$$R = \frac{TP}{TP + FN}$$

$$AP = \int_0^1 P(r) dr$$

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i$$

In the formula, AP represents the average accuracy, and P(Precision) represents the ratio between the number of samples that predict the defect category correctly and the total number of samples, reflecting the ability of the model to correctly identify the target, that is, the accuracy; R(Recall) represents the ratio of the number of samples that predict the defect category correctly to the actual total number of such samples, and represents the ability to detect the target, that is, the recall rate. In order to reflect the recognition ability of the model in all categories, the mean Average Precision (mAP) is obtained after the AP values are averaged.

3 EXPERIMENTAL RESULT AND DISCUSSION

3.1 Accuracy analysis of Pestlite-YOLO model detection

Figure 4 is a graphical comparison of detection using the Pestlite-YOLO model and the baseline algorithm at three different training epochs. It can be seen from the images that Figure 4 a-c very intuitively shows the difference and improvement in mAP, precision and recall between the original algorithm and the improved algorithm. Among all evaluation parameters, the PestLite-YOLO model is better than the baseline algorithm in testing samples. Table 1

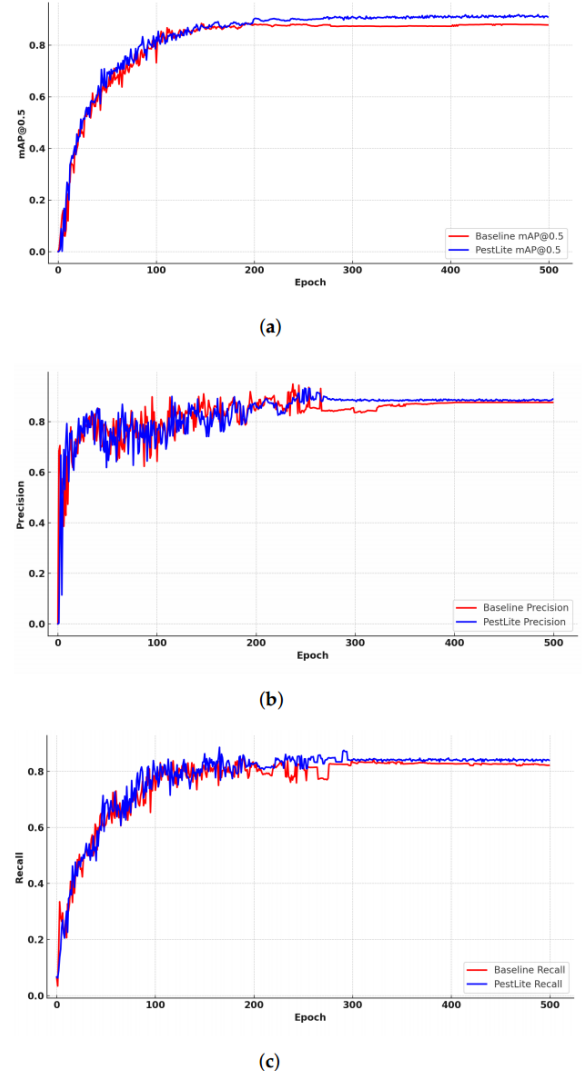


Figure 4: A graphical comparison of detection using the Pestlite-YOLO model and the baseline algorithm

shows the specific values. This further verifies the reliability and high accuracy of the detection model proposed in this paper.

As shown in Table 1, the PestLite model parameter of 6.09M proposed in this paper is significantly lower than the baseline parameter of 7.03M. The values of the Precision (P) and Recall (R) models are very similar, but the recall rate of the model proposed in this paper is significantly improved. For mAP50 (%), the PestLite model proposed in this paper is significantly improved from 86.9% to 90.8%, indicating that the accuracy of the model in identifying micro models is greatly improved.

In addition, one of the important parameters to measure the detection speed of the detection model is FPS, which specifically refers to the number of images processed per second. This model can reach 122.67 FPS. Efficiently achieves a certain balance between

Table 1: Comparison of baseline and PestLite model parameters

Model	Parameters (M)	GFLOP	P (%)	R (%)	mAP50 (%)	FPS
Baseline	7.04	15.7	91.2	81.9	86.9	108.46
PestLite	6.05	15.4	91.1	82.5	90.8	112.55

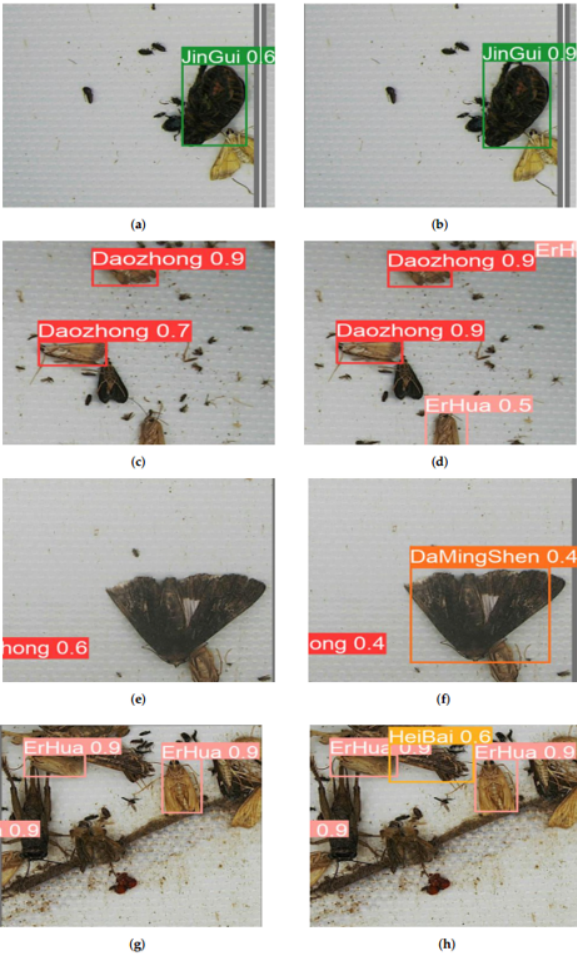


Figure 5: Comparison of traditional YOLOv5 and Pestlite-YOLO models for detecting pests

detection accuracy and speed. Making it an effective model suitable for real-time detection tasks of agricultural pest monitoring.

3.2 Visual detection effect of Pestlite-YOLO model

In order to demonstrate the actual detection effect of the model, this research work used a total of four sets of images for extensive visual comparison. The traditional YOLOv5 and the Pestlite-YOLO model proposed in this article were used to shoot and detect respectively. Among them, the crop pests captured were diverse and varied in size. As shown in Figure 5, it can be seen from the image that the model

proposed in this article has greatly improved the detection accuracy. The recognition rate of pests marked as "Golden Gui" increased from 0.6 to 0.9. At the same time, it can also more accurately identify insect types for many types of intensive pests, and has excellent multi-target detection effects.

4 CONCLUSION

Pests have caused great damage to crop yields and quality. Traditional manual inspection methods are inefficient and error-prone. Based on the YOLOv5 model, a PestLite-YOLO model for crop pest detection with multiple types and intensive detection capabilities was further developed. The model retains the backbone and head of the original model, while adding some innovative technologies and modules such as C3ECA, CARAFE, Involution and MTSPPF, especially the introduction of MTSPPF makes the model suitable for the detection of complex multi-scale pests. Involution adaptively adjusts the field of view, which enhances the detection accuracy of microscopic pests. The CARAFE operator helps to retain the feature information during the acquisition process. The above model uses a multi-level pyramid model to extract information at multiple scales. Its usefulness was extensively validated by the IP102 public pest dataset. At the same time, the mAP50 of the test sample was also greatly improved. The end result is visualized and dense and diverse areas can be detected efficiently and accurately. Insect. It is expected that practical applications will be realized in the future in crop pest and disease detection.

REFERENCES

[1] Kamilaris, A.; Prenafeta-Boldú, F.X. Deep learning in agriculture: A survey. Comput. Electron. Agric. 2018, 147, 70–90.
[2] Li, W.; Zheng, T.; Yang, Z.; Li, M.; Sun, C.; Yang, X. Classification and detection of insects from field images using deep learning for smart pest management: A systematic review. Ecol. Inform. 2021, 66, 101460.
[3] Thenmozhi, K.; Reddy, U.S. Image processing techniques for insect shape detection in field crops. In Proceedings of the 2017 International Conference on Inventive Computing and Informatics (ICICI), Coimbatore, India, 23–24 November 2017; pp. 699–704.
[4] Kujawski, A.; Vogt, K. Human Factors in Visual Quality Control. Manag. Prod. Eng. Rev. 2015, 6, 25–31.
[5] Oprea, S.; Martinez-Gonzalez, P.; Garcia-Garcia, A.; Castro-Vargas, J.A.; Orts-Escolano, S.; Garcia-Rodriguez, J.; Argyros, A. A Review on Deep Learning Techniques for Video Prediction. IEEE Trans. Pattern Anal. Mach. Intell. 2022, 44, 2806–2826.
[6] Liu, L.; Wang, R.; Xie, C.; Yang, P.; Wang, F.; Sudirman, S.; Liu, W. PestNet: An End-to-End Deep Learning Approach for Large-Scale Multi-Class Pest Detection and Classification. IEEE Access 2019, 7, 45301–45312.
[7] Redmon, J.; Divvala, S.; Girshick, R.; Farhadi, A. You Only Look Once: Unified, Real-Time Object Detection. In Proceedings of the 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Las Vegas, NV, USA, 27–30 June 2016; pp. 779–788.
[8] Xiang, Q.; Huang, X.; Huang, Z.; Chen, X.; Cheng, J.; Tang, X. Yolo-Pest: An Insect Pest Object Detection Algorithm via CAC3 Module. Sensors 2023, 23, 3221.
[9] Kim, M.; Jeong, J.; Kim, S. ECAP-YOLO: Efficient Channel Attention Pyramid YOLO for Small Object Detection in Aerial Image. Remote Sens. 2021, 13, 4851.
[10] Chen, S.; Liao, Y.; Lin, F.; Huang, B. An Improved Lightweight YOLOv5 Algorithm for Detecting Strawberry Diseases. IEEE Access 2023, 11, 54080–54092.